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IF350 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

Minggu/Pertemuan ke 8 – Scheduling and Budgeting

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Weekly Learning Outcomes of the Course (Sub-CLO)

Sub-CLO 0713 Work effectively in teams with an Agile approach and complete project deliverables on time (C5)

Learning Objectives

Objective: To analyze and apply modern techniques (2020-2025) for managing schedule and budget in complex software projects.

Key Focus Areas:
Emerging project management trends (AI, Hybrid Methods).
Best practices for time and cost estimation.
Advanced control metrics (EVM, Burn Rate).
Case study analysis of real-world budget/schedule challenges.

Target Audience: Future Software Project Managers (Focus on practical, actionable knowledge).

The Landscape 2020-2025

- AI Integration: Generative AI is increasingly used for initial estimation, resource allocation, and daily task automation (forecasting, reporting).
- Hybrid Models: Pure Waterfall is rare; most large projects use hybrid models (Agile development within a structured program framework).
- Remote/Hybrid Work: Requires new scheduling and budgeting considerations for dispersed teams and asynchronous communication.
- Data-Driven Control: Increased demand for *real-time* data (not just status reports) for stringent control and rapid decision-making.
- Value Focus: Budgets are increasingly tied to Time-to-Market (TTM) and Return on Investment (ROI), not just cost variance.

Why Schedule and Budget Control Matters

- Statistics (Post-2020): Approximately 70% of projects still fail to meet initial scope, time, *or* budget targets. (Source: Project Management Institute, various reports)

- The Iron Triangle in Software:
  Scope: Must be managed continuously (often variable in Agile).
  Time (Schedule): Directly impacts market opportunity (TTM).
  Cost (Budget): Directly impacts profitability and viability.

- Key Challenge: Balancing the need for predictability (traditional need) with the need for adaptability (modern software need).

Agile Dominance and Hybridization

Agile Methodology: The standard for software development since 2020. Focuses on iterative delivery, adapting to change, and working software over documentation.Schedule Control: Fixed duration (Sprints/Iterations), variable scope (feature flexibility).

Hybrid Project Management: The prevailing model for enterprise software. Definition: Combining a predictive approach (e.g., initial high-level planning, fixed budget) with an adaptive approach (e.g., Agile development for execution). Benefit: Provides organizational *predictability* for stakeholders while enabling team *flexibility* for developers.

Scheduling in Agile: Sprints and Velocity

Sprints/Iterations: Fixed, short timeboxes (typically 1-4 weeks).
The schedule is fixed; the scope is flexible.

Velocity: The average amount of work (usually measured in Story Points) a team completes per Sprint.
Purpose: The primary tool for forecasting future schedules.

$$\text{Remaining Sprints} = \frac{\text{Total Estimated Story Points}}{\text{Team Average Velocity}}$$

Critical Practice: Velocity must be tracked consistently (at least 3-5 sprints) before being used for reliable forecasting.

The Role of Automation and AI in Scheduling

- Generative AI (2023+ Trend): Used to generate initial task breakdowns from high-level user stories, reducing initial WBS (Work Breakdown Structure) planning time.

- Predictive Scheduling: Machine learning models analyze historical project data (past tasks, resource types, dependencies) to:
  - Suggest optimal task sequencing (beyond simple Critical Path).
  - Identify high-risk dependencies that could cause schedule slippage.
  - Benefit: Enables real-time risk adjustment of the timeline.

Impact of Remote/Hybrid Work on Schedule

- The Challenge: Time zone differences, communication lag, and loss of "osmotic communication."
- Schedule Mitigation Strategies:
  - Asynchronous-First Planning: Document decisions clearly (e.g., Confluence, Jira) rather than relying solely on meetings.
  - Overlapping Hours: Schedule critical meetings only during maximum team overlap.
  - Increased Time for Integration: Add specific "integration/merging" time at the end of sprints, as these tasks often suffer most from hybrid work fragmentation.

Practice: Timeboxing and Iterative Planning

- Timeboxing: Assigning a fixed, non-negotiable duration to an activity (e.g., a "spike" for R&D, a Sprint). Benefit: Prevents Parkinson's Law (work expanding to fill the time allotted) and forces prioritization.

- Rolling Wave Planning: Detailed planning is done only for the near-term phase (e.g., the next 2-3 Sprints), while future phases are planned at a high level. Benefit: Reduces waste from detailed planning on distant scope that is likely to change.

Zero-Based Budgeting (ZBB) in IT Projects

- Traditional: Incremental budgeting (Last year's budget + X%).
- Value-Driven Costing: Prioritizing and funding features based on theirpotential ROI (Return on Investment) or CoD (Cost of Delay).Concept: Only the highest-value features are funded first, ensuringthe biggest bang for the buck is delivered early.
- Zero-Based (ZBB): Every dollar must be justified from a "zero" base. Application: Requires the project manager to justify every cost based on its direct contribution to the project's value delivery.

Beyond Estimation: Focus on Forecasting

- Estimation (Pre-Project): Guessing the future based on past data and assumptions. (Example: COCOMO, function points).
- Forecasting (In-Progress): Calculating the future based on current performance data.
  - Crucial Metrics: EAC (Estimate At Completion) and ETC (Estimate To Complete).
  - Modern Requirement: Budget systems must update forecasts *automatically* based on real-time task completion and spend tracking.

Total Cost of Ownership (TCO) in Software Projects

- Project Budget (Initial): Focused on design, development, and testing.
- TCO (Life Cycle Cost): The full cost of acquiring, developing, operating, maintaining, and disposing of a software product over its lifespan (typically 5-10 years).
- Key Components (Post-2020):
  - Cloud Costs: Hosting, data storage, scaling fees (often a major budget overrun source).
  - Cybersecurity & Compliance: Audits, licensing for security tools.
  - Maintenance & Support: Often 15-20% of the initial development cost *per year*.
  - PM Mandate: Software PMs must now estimate and manage TCO, not just the development budget.

Budgeting in Agile (Fixed Time/Scope/Cost)

- The "Iron Triangle Flip": Agile often holds Time (Sprint Length) and Cost (Team Size) fixed, making Scope the primary variable.
- Fixed-Price Contracts: If a budget is fixed, Agile teams must manage scope through MoSCoW Prioritization (Must Have, Should Have, Could Have, Won't Have).
  - The "Won't Have" (out-of-scope) list is the primary budget defense mechanism.
- Burn Rate: The rate at which project funds are spent over time. The fundamental control mechanism for an Agile budget.

Contingency and Risk-Adjusted Budgeting

- Contingency Reserve: Funds allocated to known, accepted risks (e.g., "If integration with Legacy System X fails, we need Z budget"). Budgeting Rule: Typically 10-20% of the base estimate.

- Management Reserve: Funds allocated for *unknown-unknown* risks (e.g., a global pandemic, a sudden regulatory change). Control: Only the executive management controls this reserve.

- Risk-Adjusted Estimate: Incorporating the expected monetary value (EMV) of identified risks directly into the base estimate.

$$\text{EMV} = \text{Probability} \times \text{Impact Cost}$$

Key Budgeting Metrics

Cost Performance Index (CPI): Measures the financial efficiency of the project.

- $$\text{CPI} = \frac{\text{Earned Value (EV)}}{\text{Actual Cost (AC)}}$$

- $CPI < 1$: Over budget (Cost of work completed is greater than planned).

Budget Variance (CV): The difference between the value of work completed and the actual cost.

- $$\text{CV} = \text{Earned Value (EV)} - \text{Actual Cost (AC)}$$

To-Complete Performance Index (TCPI): The cost performance required for the remaining work to meet the original budget.

Practice: Transparent Cost Breakdown Structure (CBS)

- Requirement: A Cost Breakdown Structure must align directly with the WBS (Work Breakdown Structure).

- Modern CBS Categorization:
  - Direct Labor: Internal salaries, contractor rates, burden.
  - Indirect Costs: Project management overhead, training, facility costs.
  - Capital/Technology: Software licenses, cloud consumption, infrastructure (CaPex).
  - Risk/Contingency: Dedicated reserve funds.

- Benefit: Granular tracking allows stakeholders to see *exactly* where the money is going and enables quicker adjustments to high-spend areas.

Advanced Control & Emerging Trends

Earned Value Management (EVM) 2.0

EVM (Traditional): A powerful integration of scope, time, and cost. EV is traditionally calculated based on physical completion percentage.

Challenges in Software: How do you define "50% complete" for a piece of code?

EVM 2.0 / Fuzzy EVM:

- Adapting EVM for Agile by assigning EV based on completed Story Points (using Velocity as the baseline).
- Using predictive modeling (fuzzy logic) to estimate completion uncertainty, providing a range of possible end-dates and costs, not just a single point estimate

Data-Driven Decision Making (2020s Trend)

- Shift: Moving from anecdotal status reports ("We're on track") to objective metrics ("Our CPI is 0.92, indicating a projected 8% overrun").
- Key Data Tools: Business Intelligence (BI) and Data Visualization tools (e.g., Power BI, Tableau) are integrated with PM software (e.g., Jira, Asana).
- Mandate for PMs: Project Managers must possess strong data analysis skills to interpret metrics and present actionable insights to steering committees.

Focus on Strategic Alignment and Value (2020-2025)

Problem: Historically, projects were successful if they met *budget/schedule*, regardless of business outcome.
Modern View: A project is only successful if it meets its Strategic Business Objectives.
Metrics: Must track project metrics alongside business outcome metrics:

- Project Metric: SPI = 1.0 (On schedule).
- Business Metric: TTM reduced by 15%, Revenue increase of 5%.

Strategic Alignment: Ensures scarce resources are allocated only to projects that contribute directly to corporate goals.

Digital Tools and System Integration

- Tool Complexity: The rise of specialized tools (PMIS, CRM, Accounting, Time Tracking) necessitates integration.
- Integration Challenge: Inefficient systems lead to wasted resources, manual reconciliation of budgets/schedules, and delayed reporting.
- Best Practice: Use integrated project management software (PMIS) that acts as a single source of truth, automatically syncing time logs (schedule data) with cost rates (budget data) for real-time EVM calculation.

The Impact of Generative AI on Estimation

- Use Case: Automatically analyzing a product requirements document (PRD) or user story backlog.
- AI Functionality:
  - Compares new requirements against thousands of historical, similar tasks.
  - Generates a probabilistic estimate (e.g., 80% chance of completion in 40-50 days).
  - Suggests resource profiles based on successful past projects.
- Caveat: AI provides a powerful *baseline*, but the final estimate must still be validated and adjusted by experienced human engineers (the "Human-in-the-Loop" validation).

Key Skills for Modern Project Managers

Skill Area	Traditional Focus	2020-2025 Focus
Technical	Gantt charts, WBS creation	Data Visualization, AI Tooling, Cloud Cost Management (FinOps)
Budget	Cost Variance, Budgeting	ROI/Value Metrics, Risk-Adjusted Forecasting
Soft Skills	Command and Control	Emotional Intelligence, Asynchronous Communication, Remote Team Leadership
Methodology	Waterfall/PMP	Hybrid Frameworks, Scaling Agile (SAFe, LeSS)

TASK NEXT WEEK
Case Study: Failed IT Project (Budget & Schedule Meltdown)

The "Phoenix" System Overhaul

Project Type: Large-Scale Enterprise Resource Planning (ERP) System Replacement (Internal Software Engineering Project).

Initial Goal: Replace 20-year-old legacy systems across Finance, HR, and Supply Chain with a single, integrated platform. Initial Scope: Comprehensive, fixed requirements set upfront.
Initial Budget: $45 Million.

Initial Schedule: 24 Months.

TASK NEXT WEEK
Case Study: Failed IT Project (Budget & Schedule Meltdown)

Project Profile & Key Decisions

Parameter	Initial Plan (Waterfall/Predictive)	Reality Check
Methodology	Pure Waterfall (Requirements → Design → Build)	Hybrid (Requirements evolved constantly)
Key Resource	External Consulting Firm (Fixed Bid)	Consultant turnover was high; internal team lacked knowledge transfer.
Risk Register	Identified 50 minor risks.	Failed to predict regulatory changes that impacted core requirements.
Contingency	5% of budget ($2.25M)	Depleted within the first 12 months due to design errors.

TASK NEXT WEEK
Case Study: Failed IT Project (Budget & Schedule Meltdown)

Problem 1: Uncontrolled Scope Creep and Volatility
The Issue: Requirements were "fixed" on paper, but essential, non-negotiable legal/regulatory updates (2022/2023) forced major mid-project design changes.

Problem 2: Waterfall Rigidity vs. Complexity
The Approach: The team attempted a strict, phase-gated Waterfall approach. Testing could only begin after *all* development was complete (end of Month 24).

Problem 3: Budget Overruns and Control Failure

TASK NEXT WEEK

Case Study: Failed IT Project (Budget & Schedule Meltdown)

Problem 3: Budget Overruns and Control Failure

Metric (Month 18)	Planned Value (PV)	Actual Cost (AC)	Earned Value (EV)	Variance/Index	Analysis
Value (Cost)	$27M	$30M	$24M	**CPI = 0.80**	**$6M Over Budget.** Only 80 cents of value earned for every dollar spent.
Value (Schedule)	$27M	N/A	$24M	**SPI = 0.89**	**11% Behind Schedule.** Work completed is only 89% of work planned.

The Outcome: The Estimated At Completion (EAC), based on the current poor performance, jumped from the initial $45M to **$75M**.

TASK NEXT WEEK

Case Study: Failed IT Project (Budget & Schedule Meltdown)

The Schedule Meltdown (Delays and Dependencies)

- Cause: Critical path dependency failure. A core financial module (Activity A) missed its deadline by 3 months.
- Impact: All downstream modules (Activities B, C, D) requiring Activity A's output were halted.
- Response: Management threw more resources (people, tools) at the problem.
  - The Myth: Adding resources to a late software project often makes it *later* (Brooks' Law). New staff required training, diverting existing resources.
- Actual Schedule: Initial 24 Months-> Final 42 Months (40% delay).

Summary

- **AI & Automation:** Transitioning repetitive tasks (WBS creation, initial estimates) to AI for faster, data-driven baselines.
- **Hybridization:** Predictive governance (Budget/Release) over Adaptive execution (Sprints/Features).
- **Value-Driven:** Budget decisions are increasingly based on ROI, Time-to-Market (TTM), and Cost of Delay (CoD).
- **Control:** Increased rigor through real-time metrics (CPI, SPI, Burn Rate) and digital tool integration for transparency.
- **TCO Focus:** Project management scope is expanding to include life-cycle costs (especially cloud/operations).

Visi

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